# RECOVERY FINE-TUNING RECOVERS WHERE IT WAS TRAINED: DURATION- AND DOMAIN-DEPENDENT GAINS IN PRUNED TEXT-TO-AUDIO DIFFUSION


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## ABSTRACT

Structured pruning of text-to-audio diffusion models is followed by recovery fine-tuning, whose benefit is usually certified by one score at one inference setting. Using the released pruned and fine-tuned AudioLDM-M checkpoints at two pruning severities (65% and 83% of U-Net parameters removed), we measure the paired recovery gain in text–audio alignment across clip duration and prompt domain, anchored by a shuffled-caption chance floor and the real audio of the same prompts. At 83% pruning, fine-tuning closes 63% of the pruned checkpoint's gap to real audio at its own 10.24 s fine-tuning duration but only 33% at 3.84 s, and nothing on held-out hip-hop captions; against the unpruned model, 82% versus 44% at 83% and 52% versus 8% at 65%. The duration dependence was pre-specified, replicates on a disjoint prompt set, survives a family-wise correction, is not a scorer artefact, and is corroborated by a second scorer, event-level metrics and a frame-level grounding model. Recovery should be reported across operating points; lacking a matched dense fine-tuned control and human ratings, our claims concern evaluation, not mechanism.



Index Terms— text-to-audio generation, diffusion models, structured pruning, recovery fine-tuning, evaluation


## 1. INTRODUCTION

Structured pruning followed by recovery fine-tuning is the standard route to cheaper diffusion models [1–3], and has recently been applied to text-to-audio (TTA) generation with AudioLDM [4,5]. The recovery stage is judged by the restoration of one aggregate number—a Fréchet distance or a KL divergence on the in-domain test set at one sampler setting. A generative model, however, is used at an inference operating point: a clip duration, a prompt domain, a sampler configuration. Recovery fine-tuning is itself training at one operating point; nothing guarantees that a gain measured there holds elsewhere.

We ask: does the gain of recovery fine-tuning over the pruned checkpoint depend on the operating point at which it is evaluated? We take the released pruned and fine-tuned AudioLDM-M-Full checkpoints of Singh et al. [5] at two pruning severities, train nothing, and measure the paired recovery gain in text–audio alignment across two clip durations and two prompt domains, read against a chance floor and a real-audio ceiling. Our findings:

- Duration. The recovery gain is several times larger at the fine-tuning duration (10.24 s) than at 3.84 s: at 83% pruning it closes 63% of the gap to real audio at 10.24 s and 33% at 3.84 s (82% vs. 44% of the gap to the unpruned model; 52% vs. 8% at 65%). The interaction was pre-specified, replicates on a disjoint prompt set, and is not a scale or scorer effect (the chance floor moves by at most 0.025 between durations).
- Domain. At matched duration the gain is present in-domain (AudioCaps) and absent or negative on held-out hip-hop/rap captions, at both severities and durations; after fine-tuning, alignment on those captions sits only 0.02–0.03 above the chance floor, against 0.32 in-domain at 10.24 s.
- Where the gain lives. A frame-level grounding model shows the native-duration gain spread uniformly over the clip; a crop analysis shows the short-duration deficit arises from generating short clips, not from scoring short excerpts.

## 2. BACKGROUND

Pruning and recovery in diffusion models. Structural pruning with fine-tuning recovery is established for image diffusion models [1–3] and for self-supervised speech encoders [6, 7]. For TTA, Singh et al. [5] L1-prune the AudioLDM U-Net, fine-tune on AudioCaps for $10^6$ steps and report in-domain FAD/KL at a fixed sampler; their fine-tuned pruned model even surpasses the unpruned model in-domain (FAD 1.57 vs. 3.95), a first sign that recovery also acts as in-domain specialisation. All these works evaluate recovery at one operating point.

Evaluating text-to-audio generation. CLAP similarity [8] is the standard automatic proxy for text–audio alignment in TTA [9, 10]; FAD [11] is sensitive to sample size and embedding choice [12, 13]; event classifiers [14], human-aligned CLAP [15] and frame-level grounding models [16] measure other axes, motivating a paired, multi-metric design with one pre-specified primary endpoint.

Duration as an operating point. Latent audio diffusion conditions generation on the latent length and is trained at a fixed duration (10.24 s for AudioLDM [4, 17]; newer systems condition on timing explicitly [18]), so duration is a mechanistically motivated factor for a checkpoint fine-tuned at one duration; fine-tuning also trades in-distribution gains for out-of-distribution losses [19], motivating the domain axis.

## 3. EXPERIMENTAL DESIGN

### 3.1. Systems

The dense model is AudioLDM-M-Full [4] (U-Net stage multipliers $(1, 2, 3, 5)$, 415.96 M U-Net parameters), itself AudioCaps-fine-tuned by its authors. We study the two released pruning severities of [5]: severity 1, $(1, 2, 3, 1)$, 145.67 M parameters ($-65.0\%$), and severity 2, $(1, 2, 1, 1)$, 71.08 M ($-82.9\%$; 39% fewer multiply–accumulates [5]). At each severity we compare the pruned checkpoint (P) with the fine-tuned checkpoint (P+FT), the released result of $10^6$ steps of AudioCaps recovery fine-tuning of P [5] (EMA weights, checksums verified). Since the released pruned checkpoints carry no usable EMA weights, P is the released L1 channel selection applied to the dense EMA weights (bit-exact to the released raw weights at severity 1; at severity 2 the release differs in three decoder-seam tensors, so both conventions are evaluated: A′ primary, B′ sensitivity). "Recovery" names the fine-tuning stage, not an achieved restoration; we train no system.

### 3.2. Operating points and prompt batteries

Two durations: the native point (10.24 s, latent length 256; the fine-tuning duration) and a short point (3.84 s, latent length 96). Two domains: in-domain AudioCaps test captions [20] and held-out hip-hop/rap captions from MusicCaps [21], a sub-domain with near-zero exposure in the recovery corpus ("music" below; its captions are seven times longer than AudioCaps captions, median 56.5 vs. 8 words, so the domain axis bundles content with caption style). Severity 2 uses 192 AudioCaps prompts (disjoint from all severity-1 prompts) and 64 music prompts (3 replicates at 3.84 s, one at 10.24 s); severity 1 uses 96 AudioCaps prompts for the pre-specified domain test, a matched 80-prompt subset for the duration comparison, and 64 music prompts at 3.84 s; the dense model is generated at both durations on both AudioCaps sets (matched controls, same noise). Generation uses DDIM [22], 50 steps, guidance 2.5, $\eta = 0$, fp32, single generation, EMA weights (off the published recipe, Sec. 5).

### 3.3. Paired protocol, scoring and anchors

Within each operating point the compared systems receive identical generation noise $x_T$ per prompt (common random numbers); across durations the latent shapes differ, so duration contrasts are prompt-paired but not noise-paired. The primary scorer is fused CLAP (laion/clap-htsat-fused, rev. 365dea6e) [8], which repeat-pads 3.84 s audio to 10 s and centre-crops 10.24 s audio to 10 s; every (system, operating point) cell is scored with one fixed-order, fixed-seed call, part of the endpoint. Two anchors make absolute scores readable: the chance floor of a cell is the mean cosine between each clip and the captions of the other prompts in its battery (a shuffled-caption baseline computed from the same embeddings, so the matched scores are unchanged), and the real-audio ceiling, the real AudioCaps clip of each prompt, band-limited to the generators' 16 kHz and scored under the identical convention at its full length and as its first 3.84 s. The unit is the prompt (music replicates averaged first); intervals are 95% prompt-level percentile bootstrap intervals ($B = 10^4$).

### 3.4. Quantities and analysis plan

Let $R$ be the mean per-prompt recovery gain, CLAP(P+FT) − CLAP(P), at a given (domain, duration); $R_{\text{nat}}$, $R_{\text{short}}$ and $R_{\text{mus}}$ denote AudioCaps-native, AudioCaps-short and music-short. The duration interaction is $J = R_{\text{nat}} - R_{\text{short}} = s(P{+}FT) - s(\text{P})$, where $s(\cdot)$ is a system's duration response, its mean per-prompt score change from 3.84 to 10.24 s; $J$ is a paired per-prompt contrast with pre-specified gate $\text{lo}_{95}(J) > 0$. The recovery ratio $\rho = R/(\text{ref} - \text{P})$ is the fraction of the pruned checkpoint's gap to a reference (dense, or the real audio of the same prompts) that fine-tuning closes there. The domain contrast at matched duration is $R_{\text{short}} - R_{\text{mus}}$; the smallest effect size of interest is 0.025. "Pre-specified" means that estimand, gate and prompt set were committed to the version-controlled repository before any score was seen: the severity-1 domain test (the original reversal hypothesis; failed), the severity-1 duration follow-up, the severity-2 replication with seam sensitivity (primary), the severity-2 music cell at 10.24 s and the secondary metrics. FineLAP was frozen as a post-result diagnostic; dense control, crop, rank-scale, Holm and anchors are post-hoc.

## 4. RESULTS

### 4.1. The recovery gain grows with clip duration

Fig. 1 and Table 1 give the primary result. At severity 2, on the disjoint 192-prompt set with estimands frozen before scoring, the interaction is clearly resolved, $J = +0.159$ [+0.131, +0.187]: fine-tuning raises CLAP alignment from 0.055 to 0.299 at the native duration ($R_{\text{nat}} = +0.244$) but only from 0.015 to 0.100 at 3.84 s ($R_{\text{short}} = +0.085$; resolved, not absent); the pruned checkpoint gains just $s(\text{P}) = +0.040$ from the longer clip, the fine-tuned one $s(P{+}FT) = +0.200$; seam-robust (B′: $J = +0.161$). The anchors (Table 2) put these levels in perspective: the chance floor is close to zero and nearly duration-independent (from $-0.048$ to $+0.020$ across the AudioCaps cells; between durations it moves by at most 0.025 for any system), so $J$ is not a scale effect: on chance-corrected scores $J_c = +0.191$ [+0.162, +0.220]. Real audio of the same prompts scores 0.274 at 3.84 s and 0.440 at 10.24 s ($s(\text{real}) = +0.167$ [+0.150, +0.184], the scorer's own response to seeing the whole clip). Against this ceiling, fine-tuning closes $\rho = 63\%$ of the pruned checkpoint's gap at the native duration but 33% at 3.84 s.

At the milder severity 1 the same pattern appears with smaller effects ($R_{\text{short}} = +0.008$, $R_{\text{nat}} = +0.052$, $J = +0.044$ [−0.001, +0.087]), and the matched dense control anchors the scale. On the same 80 prompts, scored under the same convention, the dense model's duration response, $s(\text{dense}) = +0.150$ ($0.202 \to 0.352$), equals the pruned checkpoint's, $s(\text{P}) = +0.149$ (paired difference −0.001 [−0.056, +0.055]), while the fine-tuned checkpoint responds more, $s(P{+}FT) = +0.193$ (excess over dense +0.043 [−0.020, +0.109]). Part of the pruned checkpoint's raw response is a rise of its own floor (+0.019; the fine-tuned checkpoint's floor does not move), so on chance-corrected scores the severity-1 interaction is resolved, $J_c = +0.066$ [+0.020, +0.111]. In recovery ratios, fine-tuning closes 8% [−30%, +36%] of the pruned checkpoint's gap to dense at 3.84 s and 52% [+11%, +103%] at 10.24 s (Table 2). On the

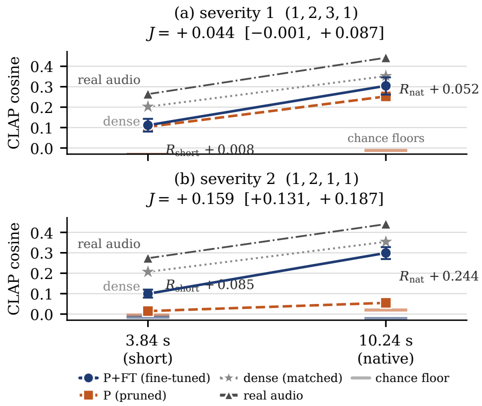


Fig. 1. The recovery gain grows with clip duration. Mean CLAP cosine of the pruned (P, dashed) and fine-tuned (P+FT, solid) checkpoints at the short and native points for (a) severity 1 and (b) severity 2; whiskers: 95% CI of the paired gain $R$, drawn about the P+FT mean; $J$ annotated. Anchors: the real audio of the same prompts (triangles, dash-dot) and each cell's shuffled-caption chance floor (short ticks). Panel (a) includes the matched dense control (stars; same 80 prompts, same scoring convention).

Table 1. Recovery gain in CLAP alignment: mean cosine per checkpoint, paired gain $R = \mathrm{CLAP}(P{+}FT) - \mathrm{CLAP}(\mathrm{P})$ with 95% CI, and fraction $W$ of prompts on which P+FT beats P. Duration-response rows give $s(\mathrm{P})$, $s(P{+}FT)$ and, in the $R$ column, $J$. †: CI excludes 0; ‡: pre-specified gate passes (Holm-corrected survivors: all severity-2 †, severity-1 music rows). $J$ paired per prompt; domain contrasts two-sample; $n$ as in Sec. 3.2.

| Setting | P | P+FT | $R$ [95% CI] | $W$ |
|---|---|---|---|---|
| Severity 1, (1, 2, 3, 1) | | | | |
| AudioCaps 3.84 s | 0.104 | 0.111 | +0.008 [-0.023, +0.039] | 0.44 |
| AudioCaps 10.24 s | 0.253 | 0.304 | +0.052† [+0.009, +0.093] | 0.64 |
| music 3.84 s | 0.117 | 0.023 | −0.094† [-0.124, -0.065] | 0.20 |
| duration $s$; $J$ | +0.149 | +0.193 | +0.044 [-0.001, +0.087] | |
| domain 3.84 s ($n$=96) | | | +0.092† [+0.054, +0.131] | |
| Severity 2, (1, 2, 1, 1) | | | | |
| AudioCaps 3.84 s | 0.015 | 0.100 | +0.085† [+0.066, +0.105] | 0.72 |
| AudioCaps 10.24 s | 0.055 | 0.299 | +0.244† [+0.215, +0.273] | 0.87 |
| music 3.84 s | 0.005 | 0.014 | +0.009 [-0.013, +0.032] | 0.48 |
| music 10.24 s | 0.089 | 0.094 | +0.005 [-0.028, +0.039] | 0.53 |
| duration $s$; $J$ | +0.040 | +0.200 | +0.159‡ [+0.131, +0.187] | |
| domain 3.84 s | | | +0.076† [+0.047, +0.105] | |
| domain 10.24 s | | | +0.239† [+0.195, +0.283] | |

Table 2. Anchors and recovery ratios (prompts and scoring as in Table 1; AC = AudioCaps): shuffled-caption chance floor of the P/P+FT cells, mean CLAP of the real audio of the same prompts, and the fraction $\rho$ of the pruned checkpoint's gap to the matched dense model and to real audio closed by fine-tuning (95% CI, in %). Dense control at severity 1: post-hoc; at severity 2: pre-specified design completion.

| Setting | floor P/P+FT | real | $\rho_{\mathrm{dense}}$ [%] | $\rho_{\mathrm{real}}$ [%] |
|---|---|---|---|---|
| sev. 1 AC 3.84 s | −0.031/ − 0.033 | 0.264 | 8% [−30, 36] | 5% [−16, 24] |
| sev. 1 AC 10.24 s | −0.012/ − 0.036 | 0.442 | 52% [11, 103] | 27% [6, 46] |
| sev. 2 AC 3.84 s | −0.005/ − 0.015 | 0.274 | 44% [36, 53] | 33% [26, 39] |
| sev. 2 AC 10.24 s | +0.020/ − 0.022 | 0.440 | 82% [72, 92] | 63% [56, 71] |
| sev. 1 music 3.84 s | +0.055/ + 0.001 | – | – | – |
| sev. 2 music 3.84 s | −0.013/ − 0.004 | – | – | – |
| sev. 2 music 10.24 s | +0.070/ + 0.061 | – | – | – |

severity-2 prompts the dense control (0.207 → 0.354) gives $s(\mathrm{dense}) = +0.147$ [+0.117, +0.177]; relative to it the pruned checkpoint's response is −0.107 [−0.137, −0.076] and the fine-tuned checkpoint's +0.053 [+0.017, +0.089] (paired), and fine-tuning closes 44% of the gap to dense at 3.84 s and 82% at 10.24 s (Table eftab:anchors).

The severity-1 raw-scale interval narrowly includes zero; the follow-up was powered for $J \geq 0.065$ at $n = 80$, so an effect of the observed size could not be resolved there. It is directionally consistent on every scale (post-hoc): the fraction $W$ of prompts where P+FT beats P rises from 0.44 to 0.64 ($\Delta W = +0.20$ [+0.06, +0.34]), the median per-prompt interaction is +0.051 [+0.012, +0.077], and the second scorer gives $J_{\mathrm{HC}} = +0.075$ [+0.012, +0.137]; at severity 2, $\Delta W = +0.15$ [+0.07, +0.22] and the median interaction is +0.172. A Holm correction over all 13 contrasts leaves every severity-2 conclusion unchanged ($p < 10^{-4}$); at severity 1, $R_{\mathrm{nat}}$ ($p = 0.016$) and $J$ ($p = 0.052$) do not survive it, so the severity-1 duration effect is established by the severity-2 replication, not on its own.

### 4.2. At matched duration the gain is domain-specific

At severity 2 and 3.84 s the gain is resolved in-domain ($R_{\mathrm{short}} = +0.085$) and null on music ($R_{\mathrm{mus}} = +0.009$; $W = 0.48$), a matched-duration domain contrast of +0.076 [+0.047, +0.105]. At severity 1 the pre-specified domain test (96 prompts) found no in-domain gain at 3.84 s ($R_{\mathrm{AC}} = -0.002$ [−0.027, +0.021]) and a negative music contrast ($R_{\mathrm{mus}} = -0.094$ [−0.124, −0.065]; domain contrast +0.092 [+0.054, +0.131]). At the native duration the severity-2 music contrast is likewise null (+0.005 [−0.028, +0.039]; $W = 0.53$), so the large native gain is confined to the fine-tuning domain (domain contrast +0.239 [+0.195, +0.283]; music duration interaction −0.004 [-0.043, +0.034]). The chance floor reads the two severities together (Table 2): after fine-tuning, alignment on the hip-hop captions is 0.022, 0.018 and 0.033 above chance (severity 1 at 3.84 s; severity 2 at 3.84 and 10.24 s), against 0.115 and 0.321 in-domain at severity 2. Whether that registers as a "penalty" depends only on whether the pruned checkpoint still had music alignment to lose: the 65%-pruned one did (0.061 above chance; chance-corrected $R_{\mathrm{mus}} = -0.040$ [−0.060, −0.020]), the 83%-pruned one did not (0.018). The null is a null near chance for both checkpoints, as CLAP measures it (no real hip-hop references were available for a ceiling), and not a caption-length effect: within AudioCaps the native gain is uncorrelated with caption length (Spearman $\rho = +0.04$ [−0.12, +0.18], severity 2); no caption is truncated by the conditioner, though 47% of the music captions exceed the 77-token length on which CLAP's text tower was pre-trained, a caption-style covariate inseparable from content.

### 4.3. Where in the clip the gain lives

Two post-result analyses locate the native-duration gain. First, as a prospectively frozen diagnostic, we score the

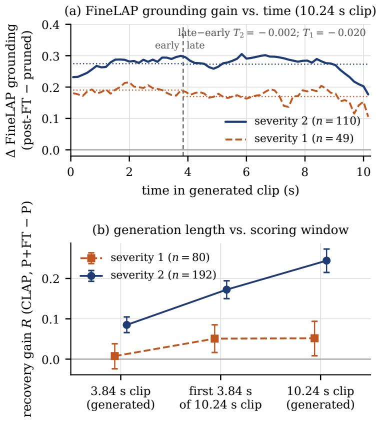


Fig. 2. (a) FineLAP frame-level grounding gain (P+FT − P) vs. time in the 10.24 s clip (dotted: window means; vertical line: 3.84 s): uniform, not back-loaded. (b) Recovery gain $R$ (95% CI) on the generated 3.84 s clip, on the first 3.84 s of the 10.24 s clip, and on the full 10.24 s clip: the crop carries the whole gain at severity 1 and over half at severity 2—a generation-length, not a scoring-window, effect.

native clips with FineLAP [16], a frame-level language–audio grounding model (EAT encoder; not CLAP-derived) reporting per 0.16 s frame how strongly the requested event is grounded (110 and 49 eligible prompts at severities 2 and 1, selected outcome-blind). Fine-tuning raises grounding across the whole clip (severity 2: semantic mass +0.27 [+0.22, +0.33]; both severities; seam-robust), and the gain in the first 3.84 s equals the gain in the remainder: $D_{\text{early}} = +0.275$ vs. $D_{\text{late}} = +0.273$, $T = -0.002$ $[-0.024, +0.020]$ (Fig. 2a). Second, a post-hoc crop analysis separates generation length from scoring window: we score only the first 3.84 s of each 10.24 s generation with the same CLAP convention and compare with the separately generated 3.84 s clips of the same prompts (Fig. 2b). The crop carries a large part of the native gain. At severity 1, $R_{\text{crop}} = +0.051$ $[+0.016, +0.085]$ equals $R_{\text{nat}}$ ($R_{\text{nat}} - R_{\text{crop}} = +0.001$ $[-0.027, +0.029]$) and exceeds $R_{\text{short}}$ by +0.043 [+0.002, +0.085]; at severity 2, $R_{\text{crop}} = +0.172$ $[+0.150, +0.194]$, $R_{\text{crop}} - R_{\text{short}} = +0.087$ $[+0.065, +0.110]$, $R_{\text{nat}} - R_{\text{crop}} = +0.072$ (seam-robust). The short-duration deficit is thus mostly a property of what the fine-tuned checkpoint generates when asked for a short clip, not of scoring a short excerpt.

### 4.4. Corroboration and pre-specified negatives

At the severity-2 native point Human-CLAP [15] (a CLAP model fine-tuned on human ratings; automatic) gives $R_{\text{nat}} = +0.375$ $[+0.340, +0.408]$ and $J = +0.185$; two pre-specified event-level metrics favour P+FT with paired CIs excluding zero: KL to real references, 2.23 vs. 4.45 ($\Delta = +2.22$ $[+1.93, +2.53]$); PANNs top-10 captured labels, 1.46 vs. 0.60 ($\Delta = +0.86$ $[+0.70, +1.02]$). FAD agrees (6.92 vs. 27.4, descriptive).

Three pre-specified hypotheses failed and are kept: (i) that recovery fine-tuning trades in-domain for out-of-domain alignment—it requires an in-domain gain at 3.84 s, and severity 1 has none ($R_{\text{AC}} \approx 0$); the loss on music is real, the trade is not; (ii) the severity-1 music penalty did not replicate (Human-CLAP alone shows −0.037 [−0.068, −0.005] at severity 2; the sign is unestablished); (iii) a "late allocation" account of the duration effect is rejected by FineLAP ($T \approx 0$). Finally, P+FT is not restored to dense: at severity 1 and 10.24 s the dense model leads P by +0.099 [+0.058, +0.140] and P+FT by +0.048 [−0.000, +0.096]; at severity 2 by +0.055 [+0.021, +0.088] at 10.24 s (+0.107 at 3.84 s).

## 5. DISCUSSION AND LIMITATIONS

Across severities, scorers and metrics, the recovery gain of the released fine-tuned AudioLDM-M checkpoints tracks the operating point of the fine-tuning itself: largest at the fine-tuning duration, confined to the fine-tuning domain, heterogeneous across prompts (column $W$). At 65% pruning the pruned checkpoint still responds to clip duration like the dense model and fine-tuning adds a modest, in-domain-only increment (8% → 52% of the dense gap). At 83% pruning the pruned checkpoint is barely above chance at either duration (0.020 and 0.035), and fine-tuning restores a duration response above the dense model's and 82% of the gap to dense—only in the fine-tuning domain and only at the fine-tuning duration (44% at 3.84 s), consistent with the recovery stage acting, at least in part, as further in-domain, native-duration specialisation. Post-pruning recovery is thus better described by a few operating-point-specific recovery ratios than by one restored score. Recovery studies should report (1) the training-matched operating point and one motivated off-native point; (2) absolute scores of P, P+FT, dense where available, and a chance floor at each point; (3) paired gains with prompt-level uncertainty under common random numbers; (4) an explicit interaction across points.

Limitations. One case study (one model family, two severities, two durations, two domains). Mechanistic attribution is blocked: the matched control—the dense model given the same AudioCaps fine-tune—was trained by the original authors but no longer exists, and the public dense text-fine-tuned release is not equivalent (different data and recipe), so the dependence cannot be attributed to pruning rather than to fine-tuning in general. Primary inference rests on CLAP-family scorers; there is no human evaluation. Sampler settings differ from the published recipe (DDIM 50 vs. 200, guidance 2.5 vs. 3.5, single vs. best-of-3; both systems evaluated identically). One held-out sub-genre (no real hip-hop references for a ceiling); cross-severity comparisons are descriptive; the severity-1 duration test was underpowered and does not survive family-wise correction; the frame-level, crop, anchor and severity-1 dense-control analyses are post-result.

## 6. CONCLUSION

Recovery fine-tuning of the released pruned AudioLDM-M checkpoints recovers much at its own operating point and much less away from it (at 83% pruning: 63% of the pruned checkpoint's gap to real audio at the fine-tuning duration, 33% at 3.84 s, none on held-out music; 82% vs. 44% of the gap to dense, and 52% vs. 8% at 65%). Audio: gbibbo.github.io/audioldm-modality-swap-pruning.

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